

Agentic Home Search & Relocation Assistant

An intelligent, multi-agent system that helps users find, compare, and evaluate homes using real listing data enriched with commute, school, safety, and lifestyle context.

POWER PUFF GIRLS TEAM

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The Problem: Housing Search Is Broken

Standard listing platforms show homes, but they don't tell you whether a home is actually a **good fit for you**. Real relocation decisions involve multiple competing priorities that are impossible to evaluate in isolation.

Budget vs. Commute

A more affordable home may mean a significantly longer daily commute, increasing hidden costs in time and fuel.

Safety vs. Price

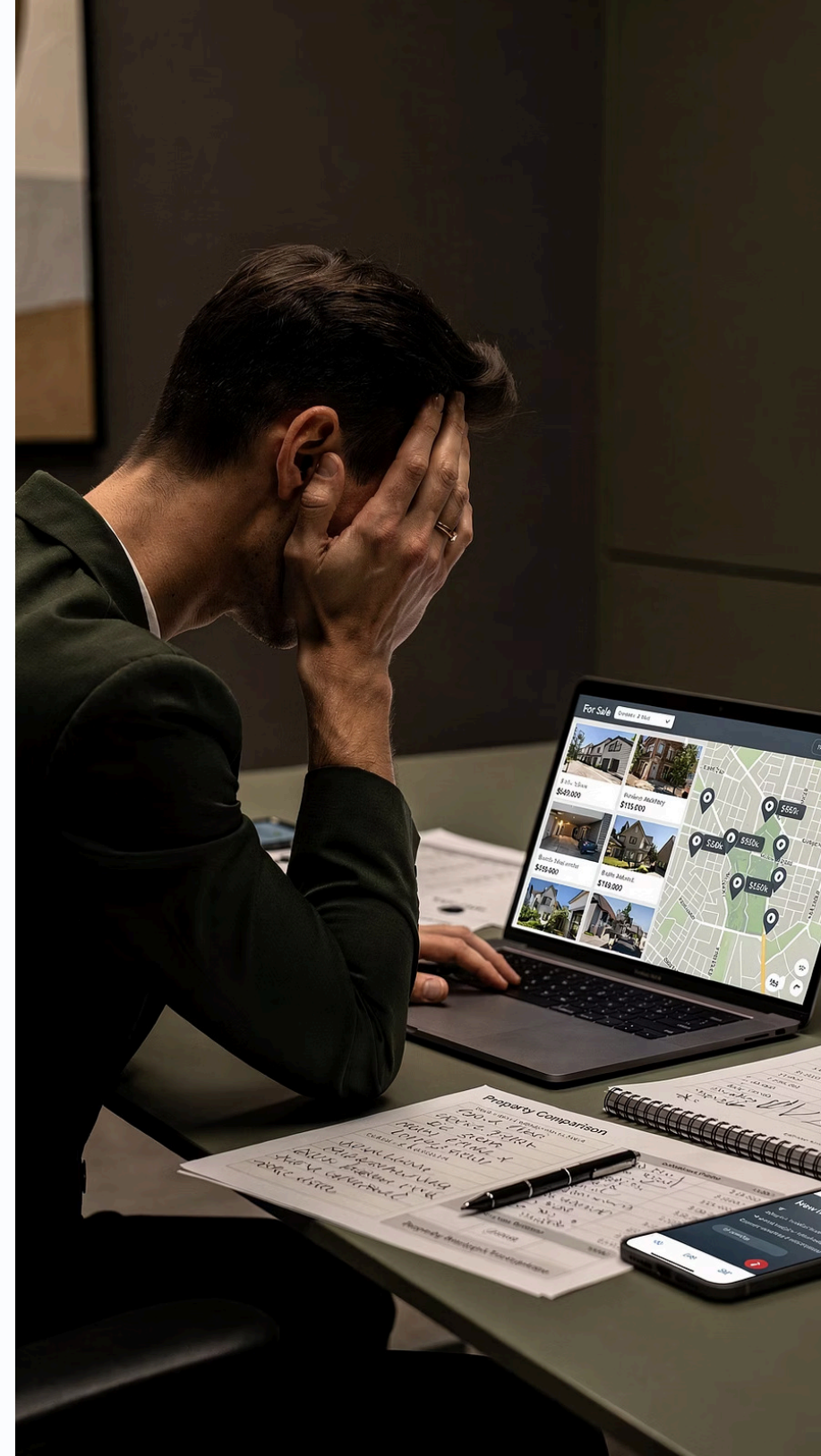
Safer neighborhoods typically command higher prices, forcing difficult tradeoffs that are hard to quantify.

Schools vs. Inventory

High-rated school districts often have limited available listings, narrowing options in ways users don't anticipate.

Lifestyle vs. Location

Walkability, parks, and grocery access vary dramatically — and are rarely surfaced in standard search filters.



Project Goal

We set out to build a system that acts not as a listing browser, but as a **relocation analyst** — one that understands multi-step goals, retrieves real data, and explains every recommendation.

1

Understand Goal

Parse complex, multi-factor user preferences into a structured relocation plan

2

Retrieve & Enrich

Fetch live listings and augment each with commute, school, and safety data

3

Rank Tradeoffs

Score and compare options against the user's specific priorities and constraints

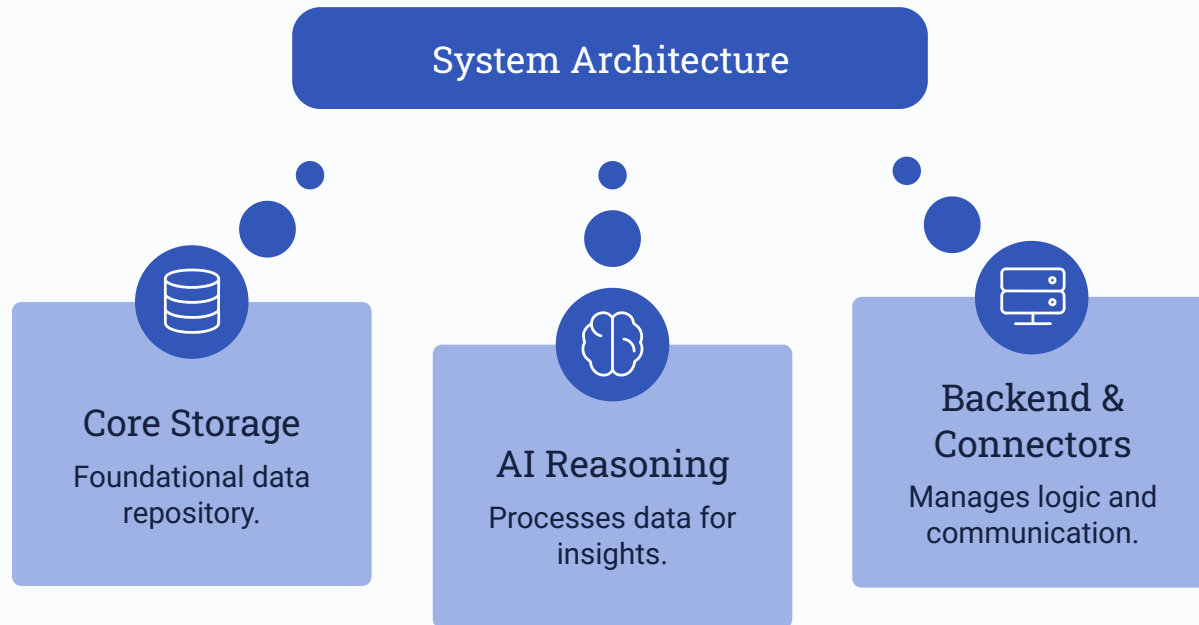
4

Explain & Critique

Deliver a grounded recommendation with transparent reasoning and identified gaps

System Architecture Overview

The platform is built on a modern, modular stack designed for reliability, scalability, and clear separation of concerns between data retrieval and AI reasoning.



The frontend collects preferences and displays results. The backend handles authentication, listing retrieval, enrichment, ranking, and admin moderation. The OpenAI layer reasons over data — it does not invent it. External providers supply ground-truth signals for listings, routes, schools, and safety.



Real Data Sources

Every recommendation is grounded in live, provider-sourced data. The AI interprets and explains — it never fabricates.



RentCast

Live sale & rental listings by ZIP, radius, price, bedrooms, and property type



OpenRouteService

Real-time commute and route estimates to workplaces and key destinations



GreatSchools

School ratings and district context for families evaluating neighborhoods



CrimeOMeter

Crime incidents, statistics, and a neighborhood safety index by location



OpenAI

Reasoning, semantic ranking, and self-critique. Interprets data — never invents it.

Agentic Workflow: Specialist Roles

The system uses a **manager-style multi-agent architecture** where a central orchestrator delegates tasks to specialized internal agents. This mirrors enterprise agent design while keeping the demo reliable and fully traceable.

Planner

- Defines the relocation task, breaks it into subtasks, and sets priorities based on user constraints.

Tool Agent

Manages provider API calls, handles retries, and flags data gaps when external calls fail.

Synthesis Agent

Writes the final recommendation in natural language, grounded in retrieved evidence.

Search Agent

Retrieves listings from RentCast and enriches each with commute, school, and safety signals.

Evaluator

Ranks listings by scoring tradeoffs against the user's weighted preferences.

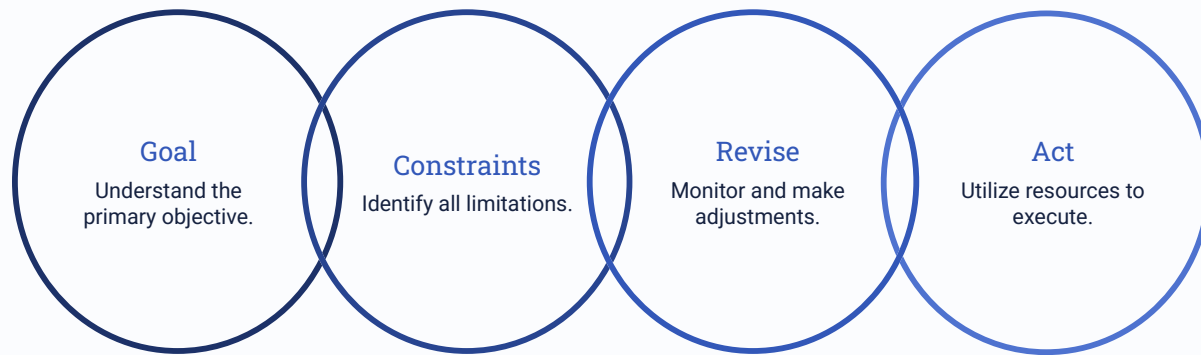
Critic & Optimizer

Reviews the answer for unsupported claims, flags missing facts, and records feedback for future improvement.



ReAct Loop: Reasoning + Acting

The agent follows a **ReAct-style loop** — a research-backed approach where the model alternates between reasoning about a problem and taking actions through external tools. This is essential for relocation, where answers cannot be generated from the prompt alone.



Why ReAct Matters Here

Relocation decisions depend on live external data. The ReAct framework allows the agent to update its plan dynamically as results come in — handling exceptions, missing data, and conflicting signals in real time.

Grounded Answers Only

No answer is produced from model knowledge alone. Every recommendation traces back to a provider call — listing data, route data, school data, or safety data — making the system auditable and trustworthy.



Self-Critique & Quality Gate

Inspired by **Self-RAG**, our system adds a built-in quality gate: after generating a recommendation, the critic agent checks every claim against retrieved evidence. If evidence is missing, the system says so — rather than presenting an incomplete answer as complete.



Retrieval

Fetch listing, commute, school, and safety data from live providers for each candidate home.



Generation

Synthesize a ranked recommendation with natural language explanations grounded in data.



Critique

Review each claim: Is it supported by listing facts? Commute evidence? School or safety data?



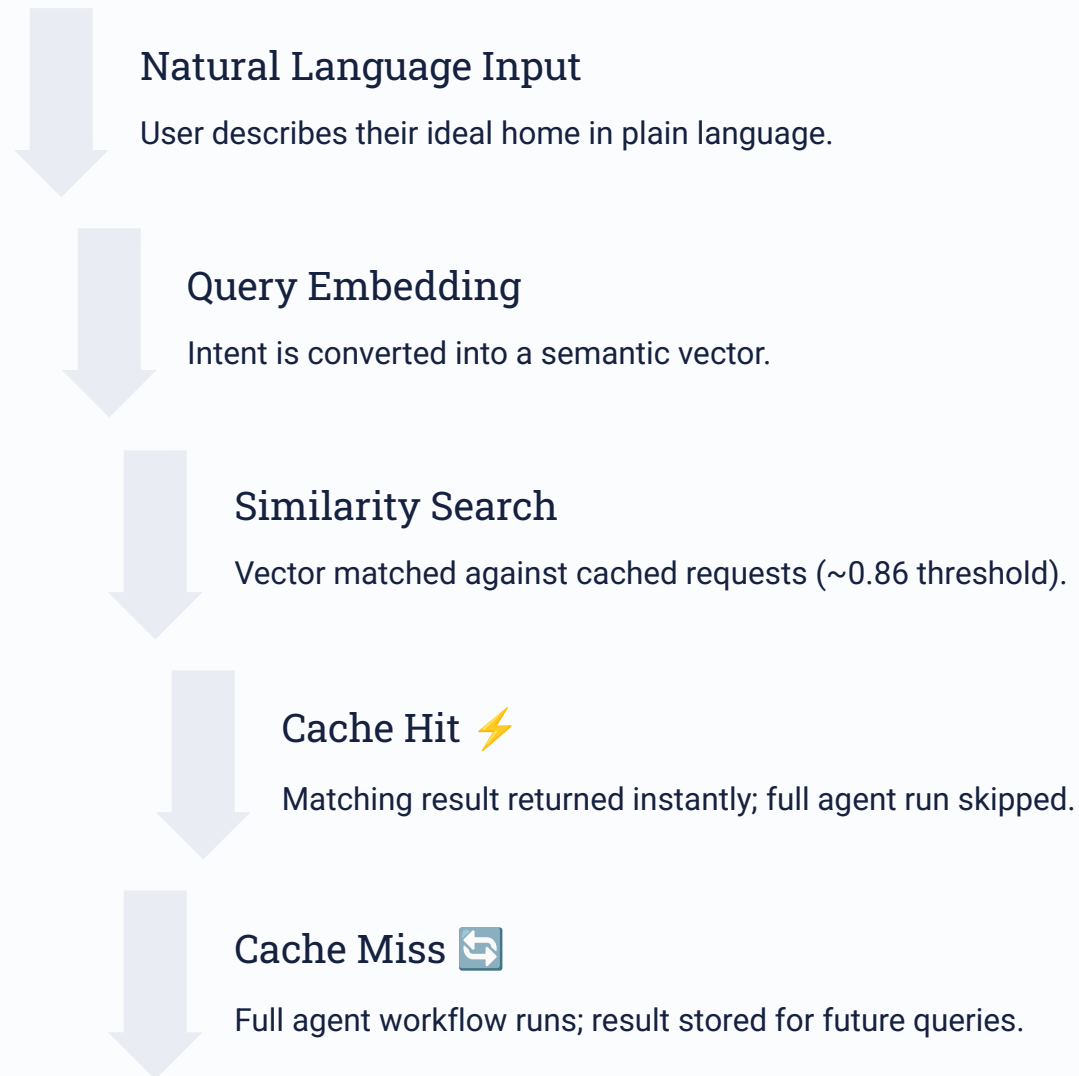
Revision

Flag unsupported claims, identify missing provider signals, and revise or caveat the answer accordingly.

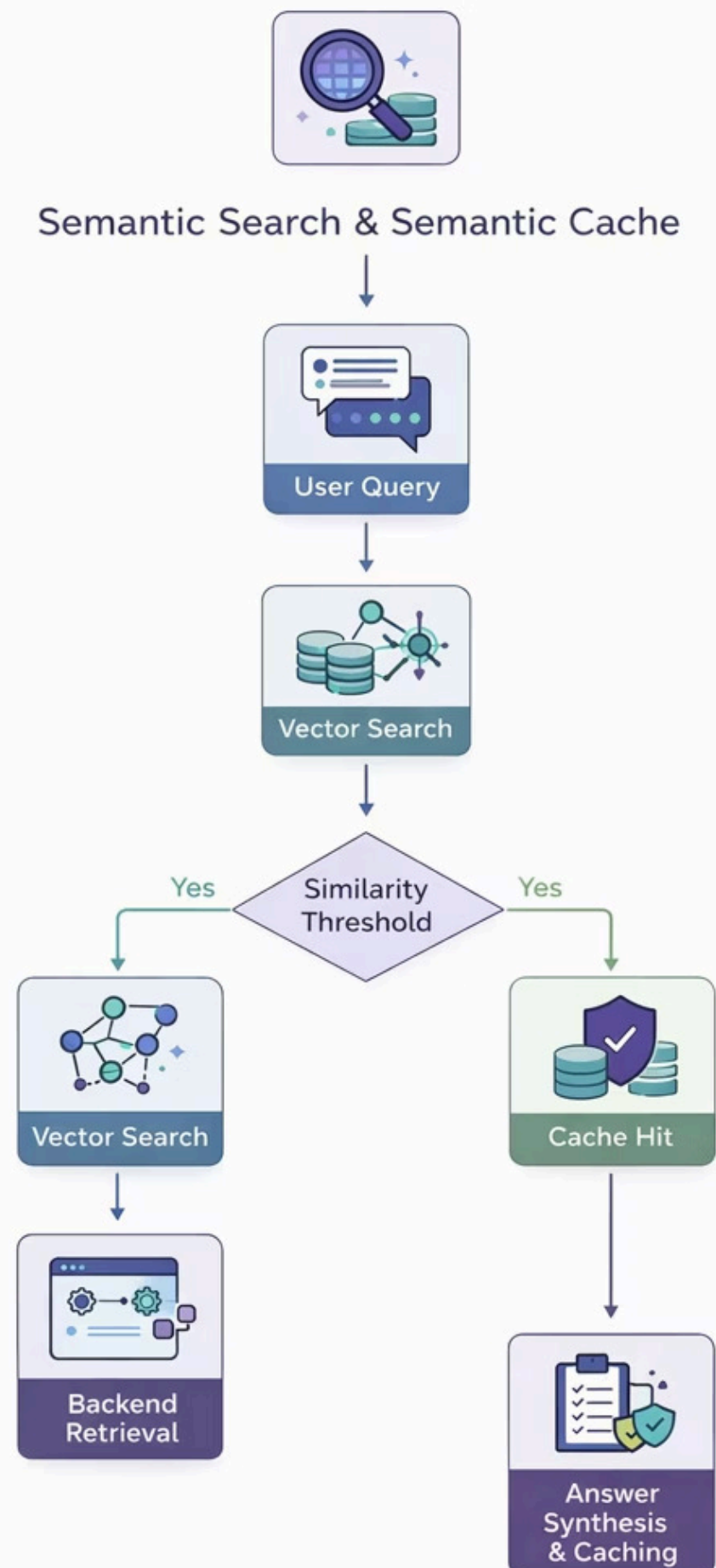
 This self-critique loop is the key differentiator from a standard search application — it improves factuality and builds user trust by being transparent about what is and is not known.

Semantic Search & Semantic Cache

Users can express intent in natural language, e.g., *"safe family relocation with strong schools and a reasonable commute."* The system embeds the query and finds the best match instantly.



❗ Exact cache is checked first. Semantic cache handles near-duplicate queries. Threshold ~0.86 balances freshness, accuracy, and cost savings.



Trust Controls & Data Validation

Real estate systems can easily mislead users when data source boundaries are unclear. Our platform enforces strict validation at every layer to ensure users always know what they are seeing.



User-Submitted Listings

Geocoded, image-checked, and range-validated before entering an admin moderation queue. Optionally reviewed by AI for anomalies.



Live Provider Data

Each record is labeled by source. Demo cache records are clearly marked and never presented as live listings.



Admin Moderation

Administrators can approve or reject advertised homes before they appear in any search results, preventing misleading listings.



How We Measure Quality

We evaluate the system from four connected angles so we can judge not only the final answer, but also how efficiently and reliably it was produced.

Recommendation Quality

We check whether the shortlisted homes truly fit the user's budget, commute, safety, and lifestyle goals. We also review whether each explanation is well supported by retrieved evidence and whether the final recommendation is clear and trustworthy.

User Experience Quality

We measure how easily users can refine constraints, understand tradeoffs, save homes, and submit listings. A strong result is not just a correct answer, but a smooth search experience that helps users make decisions faster.

Agent Workflow Quality

We track whether provider calls succeed, whether retries recover from failures, whether missing facts are detected before an answer is shown, and whether semantic caching reduces repeated work without harming correctness. This follows cache monitoring guidance emphasizing hit rate, latency saved, false positives, freshness failures, memory usage, and cost saved.

Operational Impact

We monitor search cost, cache savings, system throughput, and how often users complete the relocation task successfully. This helps us connect technical performance to real product value.

Conclusion

A relocation analyst that retrieves real data, reasons transparently, and earns user trust at every step.



Real Data Connectors

Provider-separated, live housing and neighborhood data



Semantic Search & Cache

Natural language intent matched to fast, relevant results



Transparent Agent Reasoning

Every step visible, retries handled gracefully



Self-Critique Quality Gate

Claims verified before any answer is shown



Multi-Role System

Tailored flows for users, advertisers, and admins



Thank You

Any questions?